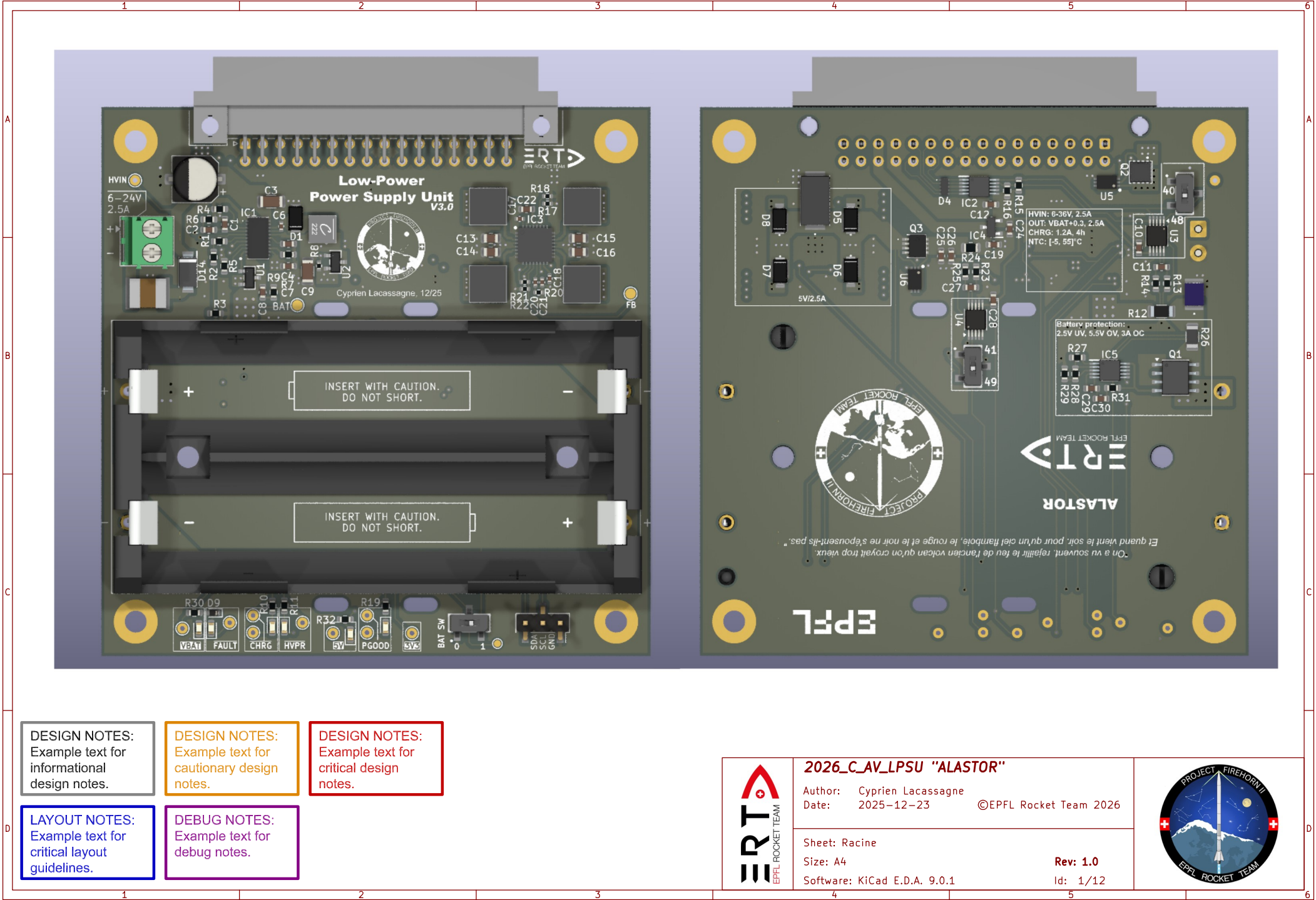
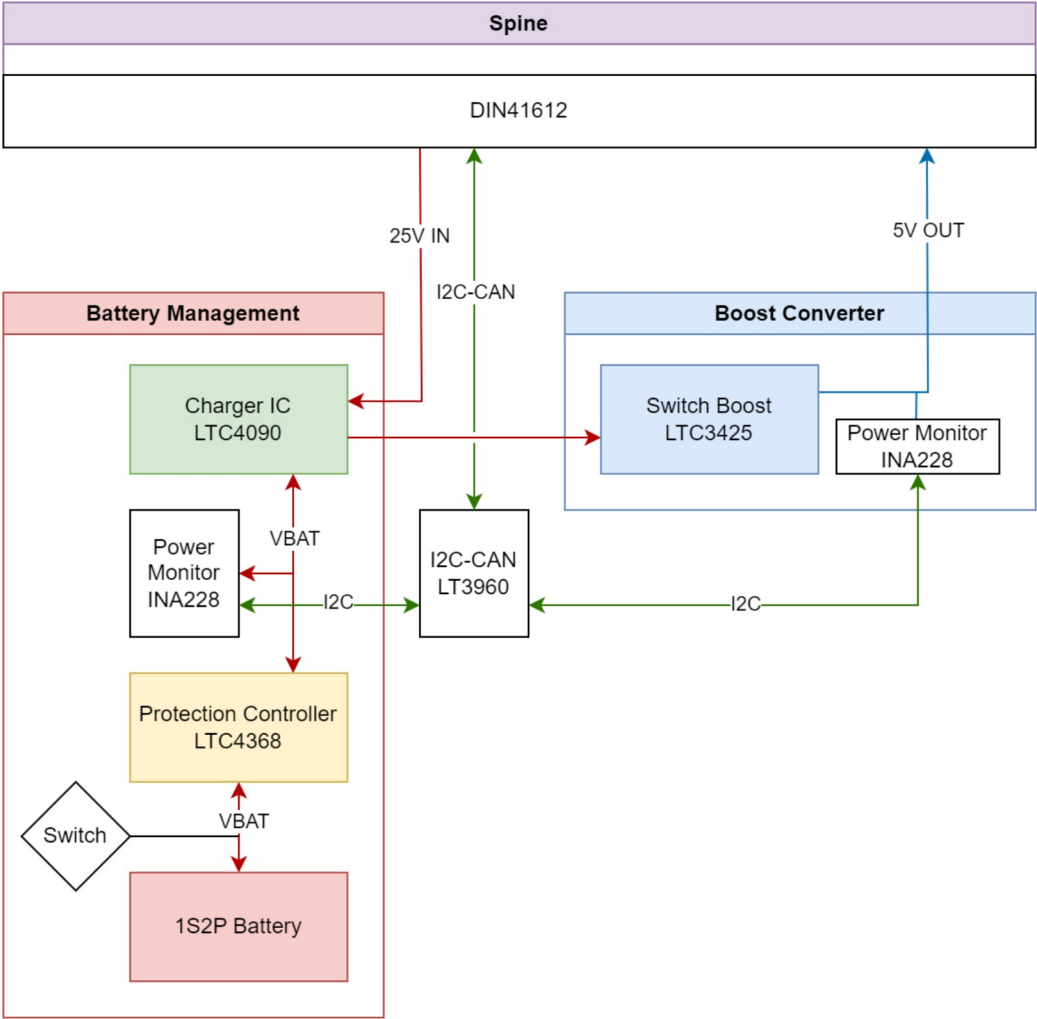
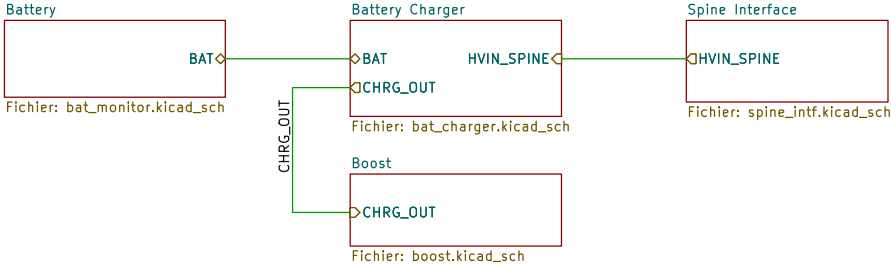




Block Diagram



Project Architecture



NAME	SOURCE	LEVEL
+++V	SOURCE NAME	??V ± ??%
++V	SOURCE NAME	??V ± ??%
+V	SOURCE NAME	??V ± ??%

Power Sequencing Diagram



2026_C_AV_LPSU "ALASTOR"

Author: Cyprien Lacassagne
Date: 2025-12-23 ©EPFL Rocket Team 2026

Sheet: Power – Sequencing
Size: A4
Software: KiCad E.D.A. 9.0.1

Rev: 1.0
Id: 4/12



[illegible][illegible][illegible][illegible]

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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

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	Author: Cyprien Lacassagne Date: 2025-12-23	©EPFL Rocket Team 2026	
	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

	2026_C_AV_LPSU "ALASTOR"		
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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

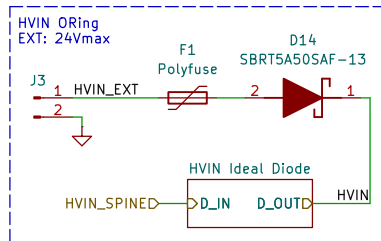
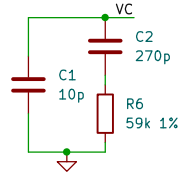
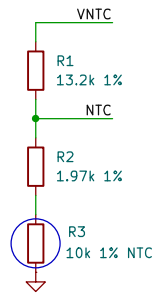
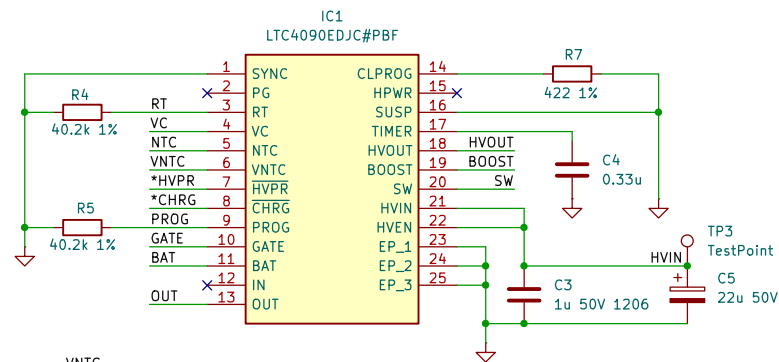
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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	

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	Sheet: Revision History Size: A4 Software: KiCad E.D.A. 9.0.1	Rev: 1.0 Id: 5/12	



Battery charger with PowerPath control
Inputs (by priority): 6V-36V DC, 3.7V Li-Ion battery
HVOUT/OUT: VBAT+0.3V if DC adapter present, else VBAT

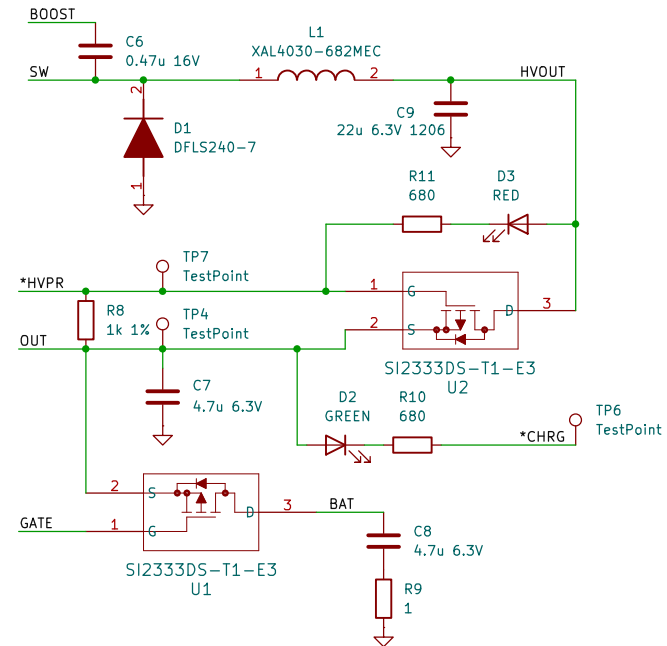
Configuration:
 $f_{OSC} = 800\text{kHz} \rightarrow R_t = 40.2\text{k}$
 $I_{IN,MAX} = 2.4\text{A} \rightarrow R_{CLPROG} = 422$
 $I_{CHG} = 1.2\text{A} \rightarrow R_{PROG} = 40.2\text{k}$
 $t_{TIMER} = 4\text{h} \rightarrow C_{TIMER} = 0.33\mu\text{F}$



BAT $\rightarrow$ BAT
OUT $\rightarrow$ CHRG_OUT

NTC config:
Charge temp. thresholds -5°C to $+55^{\circ}\text{C}$
 $r_{COLD} = 1.97\text{k}$, $r_{HOT} = 13.2\text{k}$

<https://www.vishay.com/docs/33011/convtabs.pdf>



2026_C_AV_LPSU "ALASTOR"

Author: Cyprien Lacassagne
Date: 2025-12-23 ©EPFL Rocket Team 2026

Sheet: Battery Charger

Size: A4

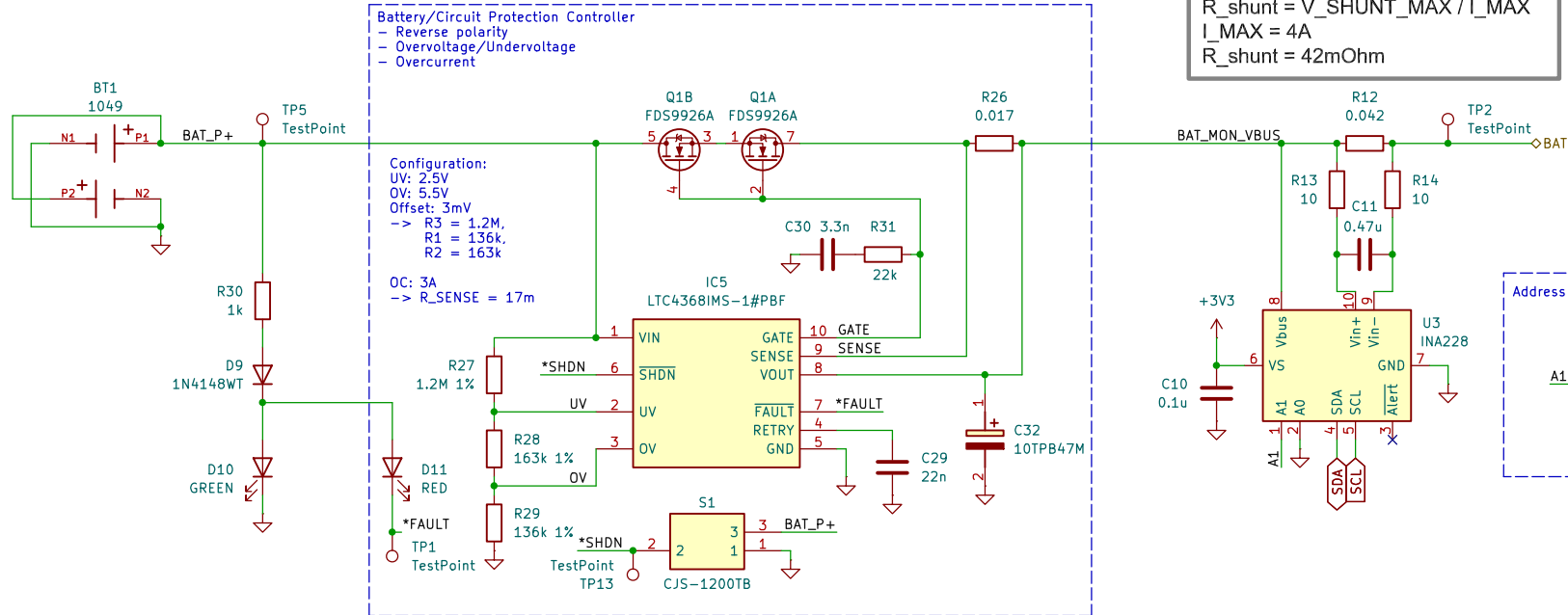
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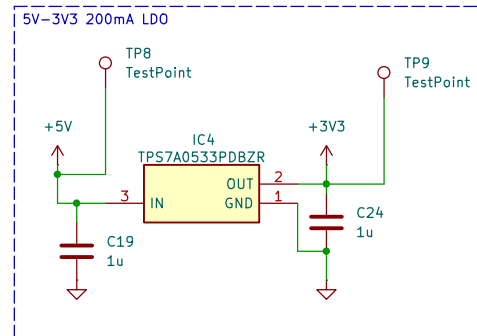
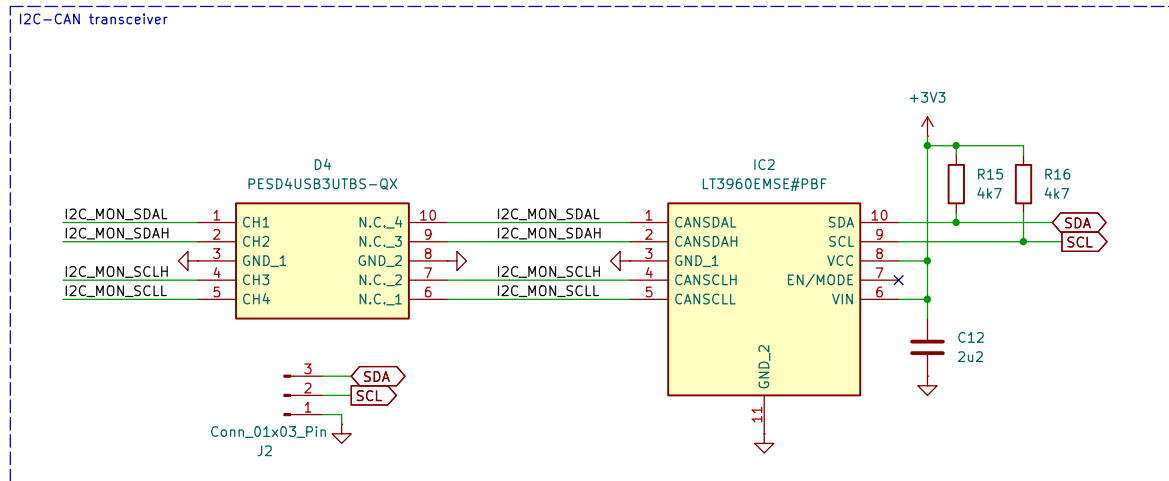
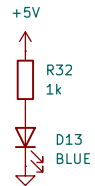
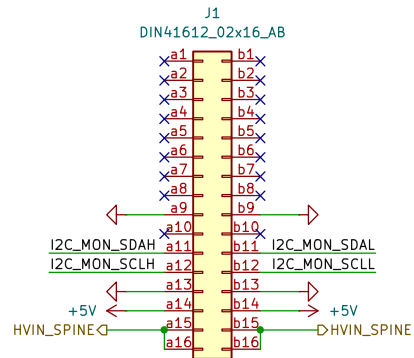
Rev: 1.0

Id: 6/12



0603L2005L-V:
 L_{hold} = 2A (1.45A at 60°C)
 L_{trip} = 4A (1s)





2026_C_AV_LPSU "ALASTOR"

Author: Cyprien Lacassagne
Date: 2025-12-23 ©EPFL Rocket Team 2026

Sheet: Spine Interface

Size: A4

Software: KiCad E.D.A. 9.0.1

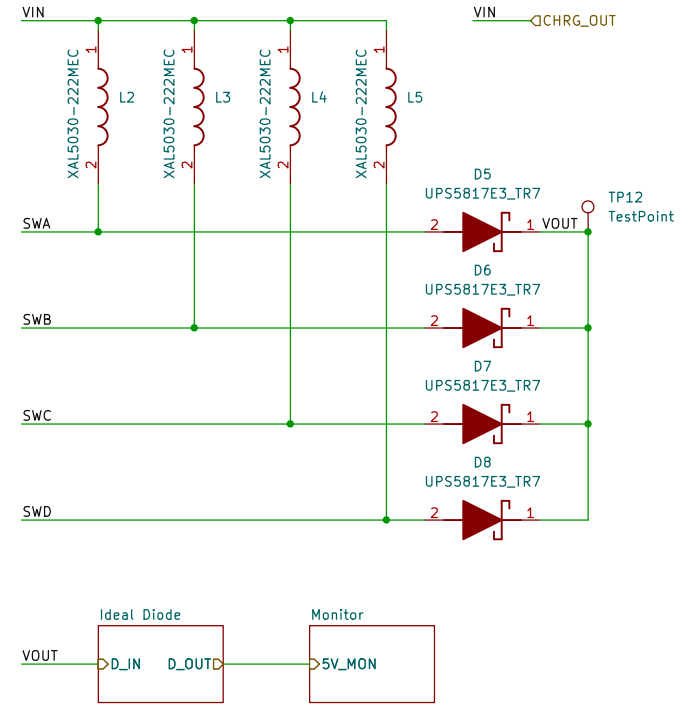
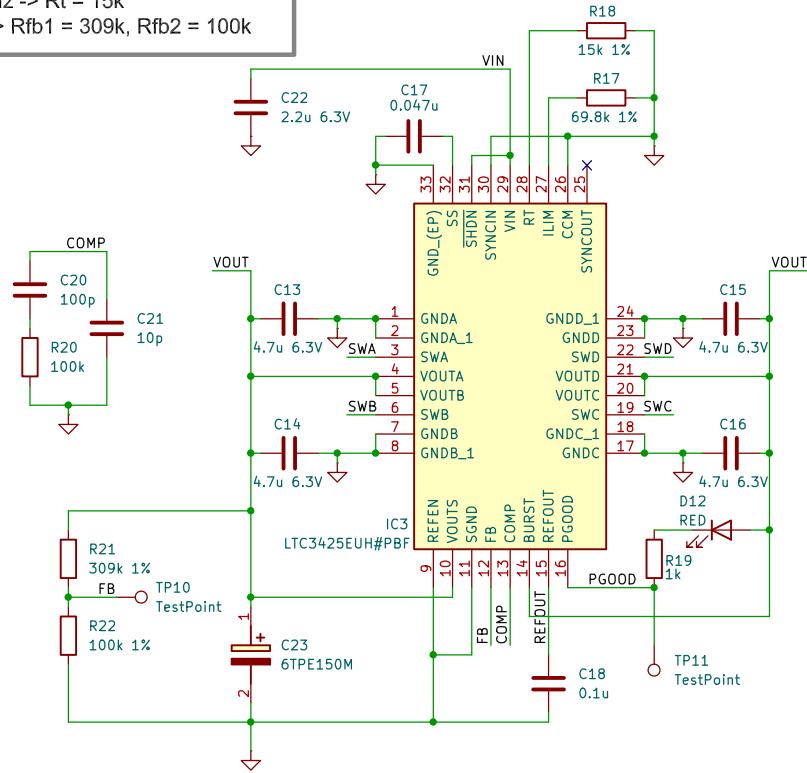
Rev: 1.0

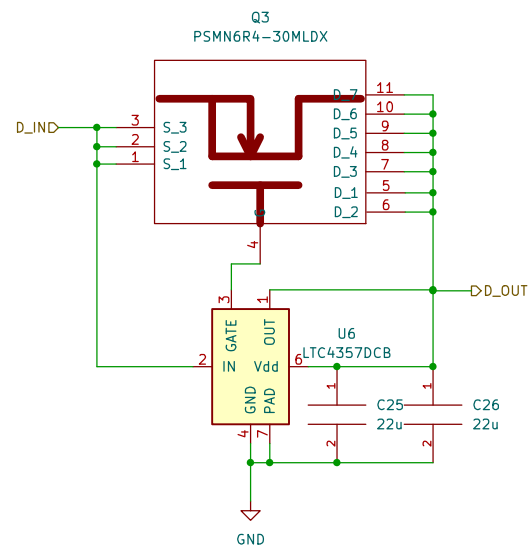
Id: 8/12



4-Phase Boost converter

Configuration:
 $f_{OSC} = 4\text{MHz} \rightarrow R_t = 15\text{k}$
 $V_{OUT} = 5\text{V} \rightarrow R_{fb1} = 309\text{k}, R_{fb2} = 100\text{k}$





2026_C_AV_LPSU "ALASTOR"

Author: Cyprien Lacassagne
Date: 2025-12-23 ©EPFL Rocket Team 2026

Sheet: Ideal Diode

Size: A4

Software: KiCad E.D.A. 9.0.1

Rev: 1.0

Id: 10/12

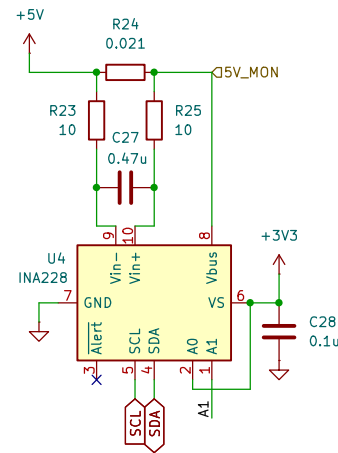
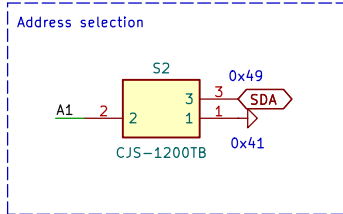


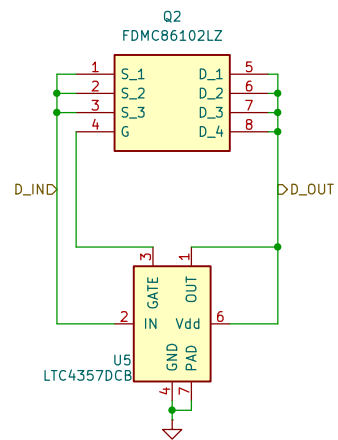
$$R_{shunt} = V_{SHUNT_MAX} / I_{MAX}$$

$$V_{SHUNT_MAX} = 163.8mV$$

$$I_{MAX} = 2A$$

$$R_{shunt} = 21m\Omega$$





2026_C_AV_LPSU "ALASTOR"

Author: Cyprien Lacassagne
Date: 2025-12-23 ©EPFL Rocket Team 2026

Sheet: HVIN Ideal Diode

Size: A4

Software: KiCad E.D.A. 9.0.1

Rev: 1.0

Id: 12/12

